

Copio

The colleague who already read everything overnight. Ask why. Get an honest written answer with citations.

Diagnostic Agent

Amazon SP-API

Phase 1.0 Dogfood

STATUS QUO

CEOs spend hours stitching data into a story

SELF-SERVE

2 to 4 hours

Brand Analytics. Helium 10. Ad console. 3PL. Swivel-chair. Biased toward the data already open.

JUNIOR ANALYST

\$60 to 90K/yr

Monday deck nobody reads. Real questions still go through Slack. Hours per question.

AGENCY

\$5 to 15K/mo

Monday memo from somebody without full SP-API access. Realtime questions go unanswered.

JUST DON'T ASK

Most common

Things slide. The cost of finding out feels too high. Compounds quietly into bad calls.

The data exists. The interpretation does not. [Copio is the interpretation.](#)

THE PROMISE

Ask why. Get a written answer.

JA James

Why are conversions down on BoCABCD123 this past week?

C copio

Conversion fell from 7.8% to 5.4% week-over-week¹, a 31% drop. Sessions held flat at 4,820². Two things broke at once: Buy Box dropped to 58% Wed³, and the 6-pack went out of stock Thu⁴.

I infer Buy Box loss is the bigger driver. I'm 70% on this.

THREE GUARANTEES

Cited. Every number links to source SP-API data with hover preview.

Calibrated. "I can see" facts read normal. "I infer" claims are *italic muted*.

Honest. When data is missing, it says so. No

HOW IT ANSWERS

Seven steps from question to written answer

1

TENANT + MEMORY

Loads LWA token, business profile, prior decisions ("don't suggest Walmart").

2

PLAN

Sonnet 4.6 decomposes the hypothesis tree. Picks 2-4 likely causes.

3

TOOL CALLS

Parallel SP-API pulls. Cache layer keyed by (tenant, query, day).

4

REASON

Marks claims as cited vs inferred. Auto-escalates to Opus 4.7 when needed.

5

DRAFT

Plain prose. Quantified uncertainty. Forbidden vocab blocked at prompt level.

6

STREAM

Token-by-token over SSE. Sub-states cycle: Planning → Fetching → Reasoning.

7

REACT

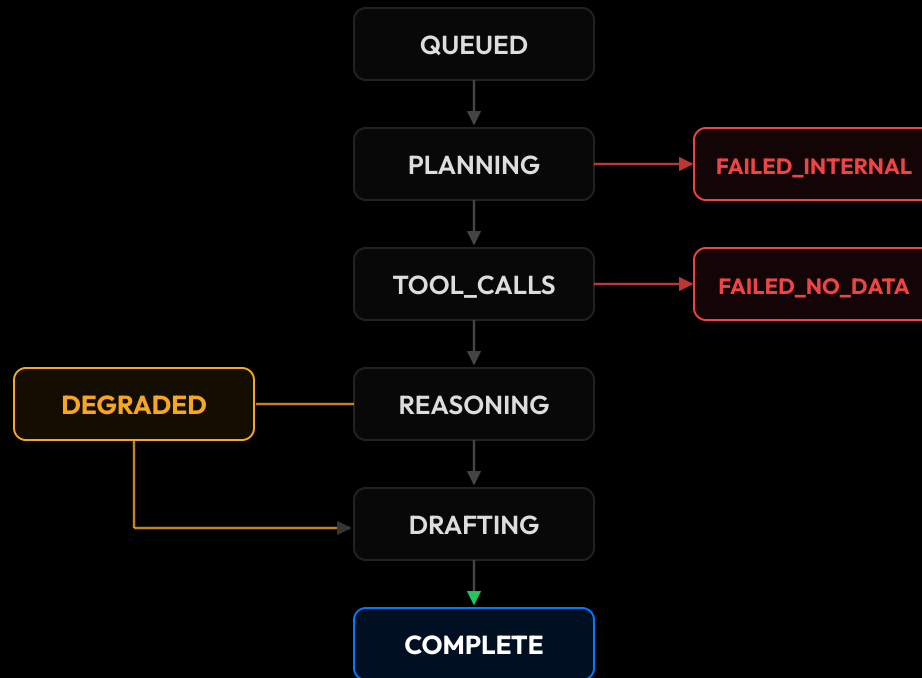
5 emojis feed the eval set. Free-text "what was wrong" is tuning signal.

∞

AUDIT LOG

Append-only. Every tool call, state change, reaction. Tier-1 ready.

The diagnostic lifecycle



INVARIANTS

Invalid transitions blocked at the orchestrator. Every state change persists to audit log.

FAILED_NO_DATA

returns honestly: "I couldn't pull anything useful right now."

DEGRADED renders a 2px cool-amber rail on

What the agent can do

 SP-API

get_orders

get_fba_inventory

get_report

Real-time + scheduled. Cached 15min hot, 24h cold. Named exceptions per Error & Rescue Map.

 MATH

summary_stats

pct_change

z_score

Anomaly detection at $|z| \geq 2$. WoW deltas. Prevents the model from inventing arithmetic.

 MEMORY

remember

recall

File-backed JSONL Phase 1.0.
Graduates to pgvector RAG in 1.1. Per-tenant scoped.

All 8 tool definitions ride the 4th cache breakpoint. Eval rig verifies every tool call returns either real data or a named degraded result.

THE LOAD-BEARING VISUAL RULE

"I can see" vs "I infer"

CITED FACT

Sales fell 8% MoM¹. Ad CPC jumped from \$0.74 to \$1.12².

INTER REGULAR

--TEXT-PRIMARY

Citation superscript points to the source SP-API response. Hover to preview the data.

INFERENCE

I infer this is competitor pressure rather than seasonality, but I'm 70% on this.

INTER ITALIC

--TEXT-MUTED

No serif italic. Differentiation via color shift + italic style. Quantifies the confidence inline.

The brand promise dies if cited facts and judgments render the same. [This is the rule the renderer enforces in voiceParser.ts.](#)

SELF-IMPROVING LOOP

Five emojis feed the eval set



Gold candidate

Promote to golden set on next nightly run.



Flag for review

Inline "tell me what was wrong" input expands. Free-text becomes tuning signal.



Exactly right

Stronger than 👍. Pinned as the answer template for that question shape.



Unclear

Voice-tuning signal. The answer was right but the prose missed.

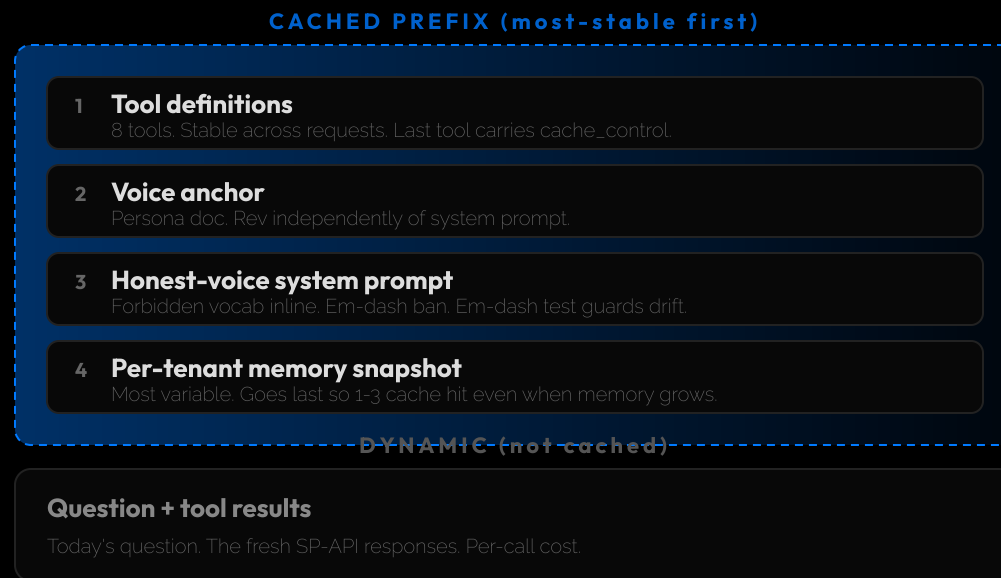


Wrong, retry

Hard regression flag. Auto-adds to the next eval-rig run.

Phase 1.0 captures the signal; the loop closes manually. Phase 1.1 wires it to a nightly Inngest cron with regression alerting.

Four explicit breakpoints, eighty percent hit rate gate



WHY EXPLICIT

Anthropic API allows
max 4 `cache_control`
markers per request.

Misplaced
breakpoints silently
degrade to **full price**.

`verify_cache_hit_
rate.py` runs cold

STACK

Locked in eng review

Backend	FastAPI · Python 3.11 · async SQLAlchemy	<i>streaming SSE + Inngest workers</i>
Database	Vanilla Postgres + pgvector (HNSW)	<i>no Supabase. Single backup target.</i>
LLM SDK	Anthropic SDK direct	<i>no LiteLLM/aisuite. caching needs explicit control</i>
Models	Sonnet 4.6 default · Opus 4.7 fallback	<i>Opus is the eval judge</i>
SP-API	python-amazon-sp-api (Saleweaver)	<i>named exceptions wrap the lib</i>
Frontend	Next.js 15 + Tailwind 4 + custom streaming hook	<i>self-hosted Inter Variable</i>
Eval rig	DIY Python, owned end-to-end	<i>the moat shouldn't be a vendor feature</i>
Audit log	Append-only Postgres table	<i>hash chain + UPDATE/DELETE block in 1.1</i>

ROADMAP

Where this goes

1.0	Single-tenant dogfood. Web chat. Polling pulls. Unnamed honest voice.	SHIPPED	internal
1.1	Multi-tenant + RLS + KMS. Named persona. Nightly eval cron. LWA OAuth onboarding.	NEXT	\$199 to \$499/mo
1.5	Slack bot. Email fallback for long diagnostics. Cross-surface memory.	PLANNED	\$199 to \$499/mo
1.6	HITL writes against Amazon (drafted PO/listing/inventory, 1-tap approve). Tier-2 ready.	PLANNED	\$299 to \$599/mo
2.0	Amazon as source of truth. Listing replication to Walmart/Shopify/eBay. Inventory sync. Order convergence.	STRATEGIC	\$499 to \$1199/mo
2.5	Morning brief. Smart silence. Decision support. Voice memo input.	VISION	\$599 to \$1299/mo

WHAT SHIPS IN PHASE 1.0

The dogfood box

DONE

14

Backend, frontend, eval rig, audit log, streaming UI, citations, reactions, honest-voice rendering, named exception map.

TESTS

32

Prompt anti-vocab, em-dash ban, all 5 emojis, math tools, memory isolation, cache placement, SP-API exceptions.

FOUNDER AUTHORIZING

5 → 20

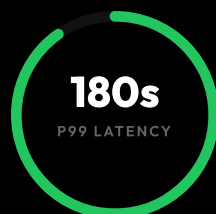
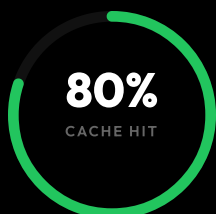
Golden questions ship as scaffold templates. Real Nutragroup ones replace before first eval run.

LOCKED FOR 1.1

8

Multi-tenant RLS, KMS, hash chain, persona name, nightly cron, reaction pipeline, dark mode, onboarding wizard.

SHIP GATES (2AM FRIDAY RULE)



All three gates fire on every release. Phase 1.1 adds: nightly judge regression alert if any case drops > 0.5 vs 7-day baseline, plus cross-tenant contamination probe.



Deep dives

Click to expand each layer

Now the founder uses it

Set ANTHROPIC_API_KEY. Connect SP-API. Author 20 real Nutragroup golden questions. Then ask the first real diagnostic and let the eval rig grow from there.

PHASE 1.0 SHIPPED

FOUNDER GATES PHASE 1.1